

Hotel Management System

Group: 1

Name	ID	Contribution
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Section: 7

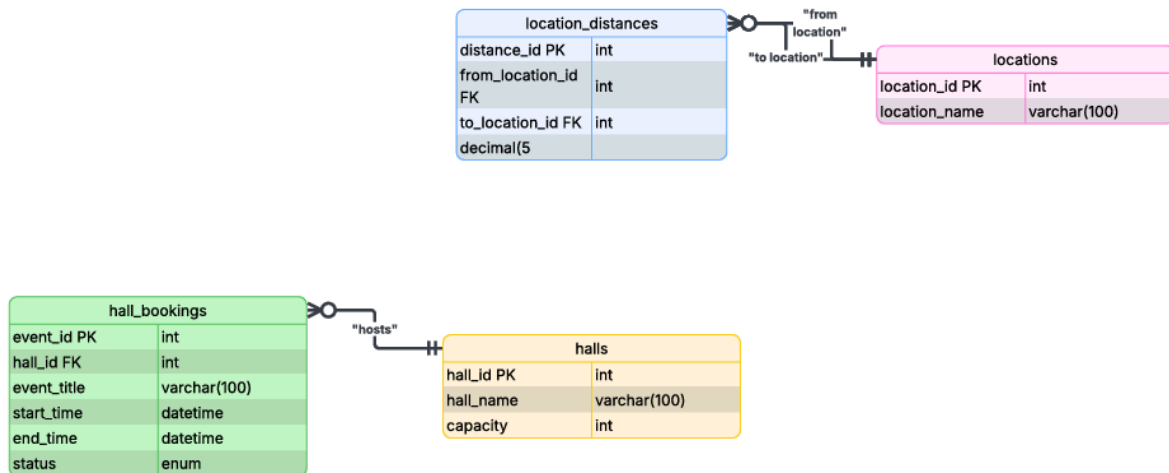
Sub: CSE302 DBMS

Semester: Summer 2026

System Stacks:

- HTML
- CSS
- JS
- PHP
- MySql (Xampp PhpMyAdmin)

Er Diagram: HD version is Included in PDF format as Separate file.



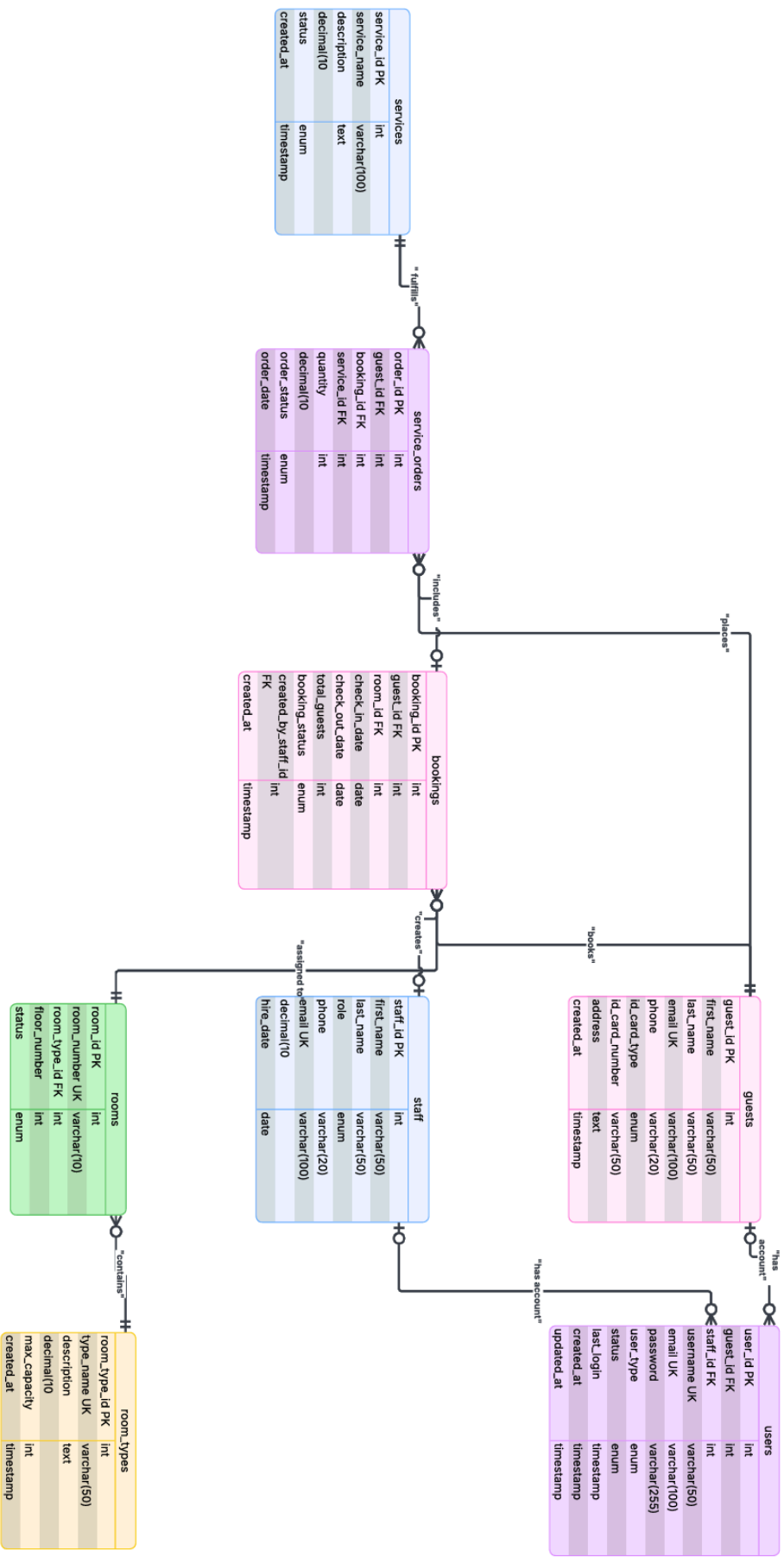


Table:

guests(guest_id, first_name, last_name, email, phone, id_card_type, id_card_number, address, created_at)

staff(staff_id, first_name, last_name, role, phone, email, salary, hire_date)

users(user_id, guest_id, staff_id, username, email, password, user_type, status, last_login, created_at, updated_at)

room_types(room_type_id, type_name, description, base_price, max_capacity, created_at)

rooms(room_id, room_number, room_type_id, floor_number, status)

bookings(booking_id, guest_id, room_id, check_in_date, check_out_date, total_guests, booking_status, created_by_staff_id, created_at)

services(service_id, service_name, description, price, status, created_at)

service_orders(order_id, guest_id, booking_id, service_id, quantity, total_cost, order_status, order_date)

halls(hall_id, hall_name, capacity)

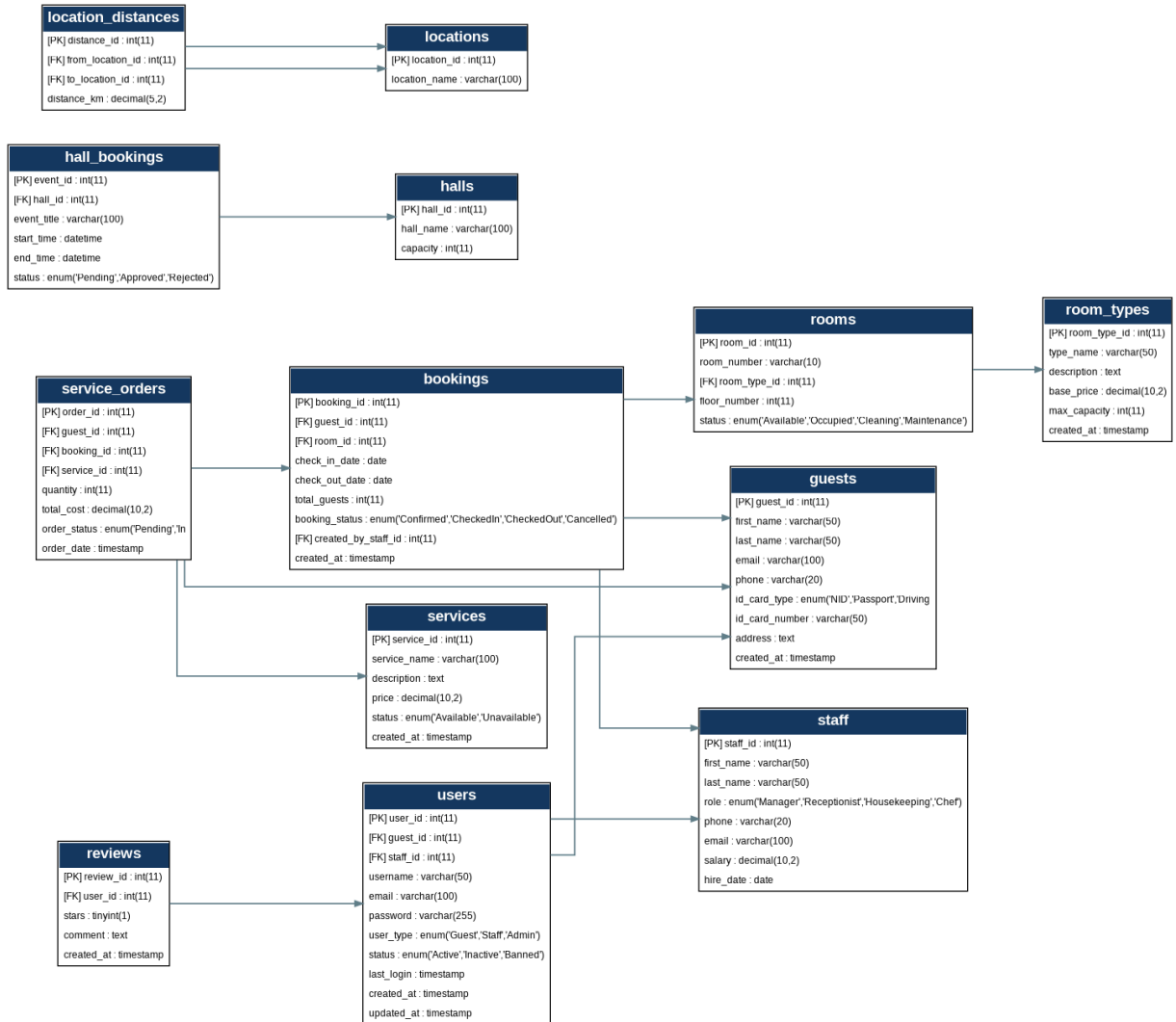
hall_bookings(event_id, hall_id, event_title, start_time, end_time, status)

locations(location_id, location_name)

location_distances(distance_id, from_location_id, to_location_id, distance_km)

reviews(review_id, user_id, stars, comment, created_at)

Relation:



Relationship summary

Parent	Child	Cardinality	Interpretation
guests	bookings	1 : many	One guest can have multiple bookings; each booking requires a guest.
rooms	bookings	1 : many	One room can appear in multiple booking records; each booking requires a room.
staff	bookings	1 : many (optional on booking)	A staff member can create many bookings; created_by_staff_id is nullable and is set to NULL on staff deletion.
halls	hall_bookings	1 : many	One hall can have multiple hall-booking events.
locations	location_distances	1 : many (from)	A location can be the source of multiple distance records.
locations	location_distances	1 : many (to)	A location can be the destination of multiple distance records.
users	reviews	1 : many	A user can submit multiple reviews.
room_types	rooms	1 : many	One room type can classify multiple rooms.
guests	service_orders	1 : many	A guest can create multiple service orders.
bookings	service_orders	1 : many (optional on order)	A booking may have multiple service orders; booking_id is nullable.
services	service_orders	1 : many	A service can appear in multiple service orders.
guests	users	1 : many (optional on user)	guest_id is nullable and not unique, so the schema does not enforce one-to-one.
staff	users	1 : many (optional on user)	staff_id is nullable and not unique, so the schema does not enforce one-to-one.

Sql Queries:

login.php

Query Type	SQL Query	Description
Single-table SELECT	SELECT * FROM users WHERE username = '\$name' AND password = '\$pass' AND user_type = '\$role' AND status = 'Active'	Authenticates user login credentials by verifying username, password, user role, and active status.
UPDATE	UPDATE users SET last_login = NOW() WHERE user_id = '{\$user_data['user_id']}'	Updates the timestamp of the user's most recent successful login.

admin_dashboard.php

Query Type	SQL Query	Description
Single-table SELECT	SELECT 1 FROM users WHERE username = '\$user_name'	Checks if a staff login username already exists prior to creating an account.
INSERT	INSERT INTO staff (first_name, last_name, role, phone, email, salary, hire_date) VALUES (...)	Inserts a new staff member profile into the system.
INSERT	INSERT INTO users (staff_id, username, email, password, user_type) VALUES (...)	Creates a staff user account linked to the new staff ID.
Single-table SELECT	SELECT 1 FROM room_types WHERE type_name = '\$type_name'	Validates whether a room category name already exists.
INSERT	INSERT INTO room_types (type_name, description, base_price, max_capacity) VALUES (...)	Adds a new room category along with pricing and capacity constraints.
Single-table SELECT	SELECT 1 FROM rooms WHERE room_number = '\$room_number'	Ensures duplicate room numbers are not added.
INSERT	INSERT INTO rooms (room_number, room_type_id, floor_number, status) VALUES (...)	Registers a new hotel room assigned to a specific room category.
Single-table SELECT	SELECT 1 FROM locations WHERE location_name = '\$location_name'	Verifies if a map location node already exists.
INSERT	INSERT INTO locations (location_name) VALUES ('\$location_name')	Inserts a new map location node into the database.
INSERT	INSERT INTO location_distances (from_location_id, to_location_id, distance_km) VALUES (...)	Saves distance edge connections between two locations.
INSERT	INSERT INTO halls (hall_name, capacity) VALUES ('\$hall_name', '\$capacity')	Registers a new banquet or conference hall.
INSERT	INSERT INTO services (service_name, description, price, status) VALUES (...)	Creates a new payable guest service offering.
Single-table SELECT	SELECT room_type_id, type_name, base_price FROM room_types	Populates category selection options when creating a room.

Query Type	SQL Query	Description
View SELECT	SELECT * FROM average_income_per_service	Fetches aggregated financial analytics from the average income database view.
Single-table SELECT	SELECT location_id, location_name FROM locations	Populates source and destination dropdowns for distance mapping.

staff_dashboard.php

Query Type	SQL Query	Description
Single-table SELECT	SELECT role FROM staff WHERE staff_id = '\$staff_id'	Fetches the job role of the active staff session to enforce permissions.
UPDATE	UPDATE service_orders SET order_status = '\$new_status' WHERE order_id = '\$order_id'	Updates the order status of a requested room service item.
Single-table SELECT	SELECT guest_id, first_name, last_name, email, phone, id_card_type, id_card_number FROM guests	Retrieves guest records to perform exact-match linear search checks.
Single-table SELECT	SELECT 1 FROM users WHERE username = '\$guest_username'	Validates username availability for guest user registration.
INSERT	INSERT INTO guests (first_name, last_name, email, phone, id_card_type, id_card_number, address) VALUES (...)	Registers personal details of a walk-in guest.
INSERT	INSERT INTO users (guest_id, username, email, password, user_type) VALUES (...)	Creates a guest user login account.
INSERT	INSERT INTO bookings (guest_id, room_id, check_in_date, check_out_date, total_guests, booking_status, created_by_staff_id, created_at) VALUES (...)	Creates a new booking reservation for a guest.
UPDATE	UPDATE rooms SET status = 'Occupied' WHERE room_id = '\$room_id'	Marks a room's status as occupied upon check-in.
Single-table SELECT	SELECT guest_id FROM guests WHERE LOWER(first_name) = LOWER(...) AND ...	Searches for an existing guest profile matching exact identity criteria.
Single-table SELECT	SELECT room_id FROM bookings WHERE booking_id = '\$booking_id' AND booking_status IN ('Confirmed', 'CheckedIn')	Validates active booking status before executing check-out.
UPDATE	UPDATE bookings SET booking_status = 'CheckedOut' WHERE booking_id = '\$booking_id'	Marks a booking as completed during guest check-out.
UPDATE	UPDATE rooms SET status = 'Available' WHERE room_id = '\$room_id'	Releases an occupied room back to available status.
Single-table Aggregate SELECT	SELECT SUM(total_cost) AS total_service_fee FROM service_orders WHERE booking_id = '\$booking_id' AND order_status = 'Completed'	Sums up all completed service charges incurred during a guest stay.
INSERT	INSERT INTO hall_bookings (hall_id, event_title, start_time, end_time, status) VALUES (...)	Submits a new event booking request for a banquet hall.
Single-table SELECT	SELECT * FROM guests	Retrieves all guest records for linear multi-criteria guest search.

Query Type	SQL Query	Description
Multi-table SELECT	SELECT r.room_id, r.room_number, rt.type_name, rt.base_price, rt.max_capacity FROM rooms r JOIN room_types rt ON r.room_type_id = rt.room_type_id WHERE r.status = 'Available'	Retrieves available rooms joined with pricing and capacity data.
Multi-table / Subquery SELECT	SELECT b.booking_id, g.first_name, g.last_name, r.room_number, (SELECT IFNULL(SUM(so.total_cost), 0) FROM service_orders so WHERE ...) AS completed_service_total FROM bookings b JOIN guests g ON ... JOIN rooms r ON ... WHERE b.booking_status IN ('Confirmed', 'CheckedIn')	Displays active bookings joined with completed service totals for billing.
Multi-table SELECT	SELECT b.booking_id, g.first_name, g.last_name, r.room_number, b.total_guests, b.check_in_date, b.check_out_date, b.booking_status FROM bookings b JOIN guests g ON b.guest_id = g.guest_id JOIN rooms r ON b.room_id = r.room_id WHERE b.booking_status IN ('Confirmed', 'CheckedIn')	Summarizes active room reservations.
View SELECT	SELECT * FROM pending_service_orders ORDER BY order_id DESC	Fetches pending room service requests from the database view.
Multi-table SELECT	SELECT r.room_number, rt.type_name, r.floor_number, r.status FROM rooms r JOIN room_types rt ON r.room_type_id = rt.room_type_id ORDER BY r.room_number ASC	Displays overall room inventory status and associated categories.
Single-table SELECT	SELECT hall_id, hall_name, capacity FROM halls	Lists all halls for event scheduling options.

Php Basics:

In order to fetch data from PhpMyAdmin sql we used `mysqli_query` and `mysqli_fetch_assoc`

Example

```
<?php
    $sql = "SELECT * FROM room_types";
    $result = mysqli_query($conn, $sql);
    while ($row = mysqli_fetch_assoc($result)) {
        ?>
    HTML DIVs
<?php } ?>
```

Here it will keep retrieving data from until the row is empty

We also used `$_Session` to keep track of user profile. In order to get data from the front end we used the `IS_SET` method with post or get way.

```
$is_logged_in = isset($_SESSION['username']);
$user_name = $is_logged_in ? $_SESSION['username'] : '';
$user_type = $is_logged_in ? $_SESSION['user_type'] : '';
```

Authorization data:

*Note: in order to use these you must first import **ahmani_hotel (1).sql** inside phpMyAdmin first also change port inside db.php*

Login as Guest:

Name: abir

Pass: 123

Login as Staff:

Name: abcd

Pass: 123

Login as Administrator:

Name: admin

Pass: admin

Concluding Remarks

We made this project entirely by hand altho we used help of Ai during making of CSS file but the basic CSS was made by me but few part like golden glow and complex style was taken from Ai's help we used Gemini Basic Chat. We could also add a few parts like in admin we could add update or modification of data feature but it would have made the Project way more complex and way larger for a single subject mini project as it is already a project with more than 7000+ lines of code. This project was actually hard and it took 10 days with Me and Nafis working together while also maintaining quiz and presentation of other subjects. I will be a bit biased here, but some other project maybe better looking or modern finish then our one but I highly doubt if they actually made that project entirely by their own knowledge without any help of Ai in core project building. We used almost basic structure for out project.